

PENGWIN 2026 Task 1 — V2.6 Updated-Data STU-Net Submission

Algorithm Description · 2026-07-24

1. Submission summary

This submission performs per-fragment instance segmentation of pelvic and femoral fractures in CT. It uses a two-stage STU-Net-B cascade:

1. **Stage A — anatomy segmentation:** a whole-volume five-class network predicts background, sacrum, left hip, right hip, and femur.
2. **Official case routing:** the pelvic/femur decision functions published by the PENGWIN organizers select the target anatomy family.
3. **Stage B — fracture segmentation:** a CT-only network predicts four ABBC channels plus nine learned same-instance affinity channels for each target-anatomy ROI.
4. **Instance decoding and assembly:** average-linkage affinity agglomeration produces fragment instances, removes components below approximately 1 cm³, and maps the instances to the official PENGWIN label ranges.

V2.6 keeps the verified inference code, official case router, network architectures, decoder, thresholds, and post-processing unchanged. Both deployed checkpoints were retrained after rebuilding the datasets from the refreshed official PENGWIN release:

- Stage A: PenguinTrainerSTUNetBaseAnatomyV301, fold 0
- Stage B: PenguinTrainerSTUNetBaseAffinityV308, fold 0

2. Data and split

Training uses the official PENGWIN 2026 Task 1 release containing 340 CT cases: 170 pelvic-fracture cases and 170 femoral-fracture cases. A case-grouped five-fold split with seed 12345 is shared by both stages. The deployed fold-0 models are trained on 272 cases and evaluated locally on the same held-out 68 cases (34 pelvic and 34 femoral) without patient overlap between training and validation.

Stage A uses a partial-label-aware anatomy target: pelvic cases supervise sacrum and left/right hips, while femoral cases supervise femur. A marginal Dice+cross-entropy loss prevents unlabeled anatomies from being treated as true background.

Stage B uses one CT channel only. The Stage-A probability is used to localize the ROI but is not provided as a Stage-B input, avoiding cascade feature leakage.

3. Model architecture

Both stages use STU-Net-B, a 3D residual U-Net with approximately 58 million parameters. The networks are initialized from a publicly available TotalSegmentator bone pretrain. Stage A has a five-class softmax head. Stage B emits 13 channels:

- four ABBC logits: background, outer border, inter-fragment boundary, and core;

- nine affinity logits at short, middle, and long offsets along the three spatial axes.

The affinity loss is class-balanced so that rare cross-fragment edges are not overwhelmed by the more frequent same-fragment voxel pairs.

4. Routing and decoding

The container calls the organizers' published `get_image_info()` and `classify_pelvic_femur()` rules using the specified SimpleITK/NumPy axis mapping. Pelvic cases process Sacrum, LeftHip, and RightHip. Femur cases process Femur only. The legacy random-forest router is disabled and no router artifact is required.

For every routed anatomy, Stage A supplies a padded ROI. Stage B predicts the ABBC and affinity fields in that ROI. The decoder first oversegments the foreground at learned separation ridges, then merges adjacent supervoxels using average-linkage interface affinity with `PENGWIN_AGGL0_T=0.45`. Small components are pruned and local instance IDs are mapped to:

- sacrum: 1–50
- left hip: 51–100
- right hip: 101–150
- femur: 151–200

5. Local validation

The end-to-end cascade was rerun on all 68 held-out fold-0 cases with the updated Stage A and Stage B checkpoints. The project's official-aligned local proxy produced:

Metric	V2.6 updated-data checkpoints
Fracture IoU	0.847755
Fracture Dice	0.887390
Local Dice (20 mm)	0.887545
HD95	3.630712 mm
ASSD	0.930238 mm
Instance Recall	0.949466
Instance Precision	0.910110
Instance F1	0.914869
Topology Consistency	0.295903
Merge Errors	10
Split Errors	500

These are local proxy values, not results from the unpublished Grand Challenge test evaluator. They are reported for reproducibility and model-version verification.

6. Deployment contract

The algorithm is packaged as a non-root Docker container based on `pytorch/pytorch:2.1.2-cuda11.8-cudnn8-runtime`. Grand Challenge extracts the separate model archive under `/opt/ml/model`. The container loads Stage A and Stage B serially, so peak GPU memory is the maximum of the two stages rather than their sum. The runtime is designed for an NVIDIA T4 16 GiB GPU and the 10-minute per-case limit.

The container logs the network class and a weight checksum at startup. The V2.6 checkpoints produce approximately:

- Stage A w0sum: 1.0411e+02
- Stage B w0sum: 1.0841e+02

7. Checkpoint identity

SHA-256:

- Stage A V301: 0d52f0fa41a69462d9ff757fb9417e70d1101104a896e5f4ed709b4ea2566509
- Stage B V308: 66c1b47d9df250add49bff9997c373be949653a6f9651a281081be896d53534c
- Grand Challenge model archive model_v301_new_v308_new_20260724.tar.gz:
7ba9fa8e6b6ac95bbfcdc006573c67e915d993466019abadb18da0cf36a4d240

8. Limitations

- The official hidden-test evaluator is unavailable locally.
- This is a single-fold, single-seed submission without ensembling.
- The refreshed models improve recall and surface-distance metrics locally but reduce precision and instance F1 relative to the previous checkpoints.

9. References

- Isensee F. et al. nnU-Net. *Nature Methods* 18, 203–211 (2021).
- Huang Y. et al. STU-Net.
- PENGWIN 2026 Task 1 official baseline.
- PENGWIN 2024 first-place ABBC formulation.
- Bailoni A. et al. GASP average-linkage agglomeration, CVPR 2022.